

DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning**



INTERNSHIP REPORT

ON

“DATA INTERN”

Submitted in partial fulfillment for the award of degree(21INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Nisarga B - 1DS21AI037

Conducted at



Quince

Flipping The Way We Shop

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**Data Intern**” carried out by **Ms NISARGA B [IDS21AI037]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

**Signature of
Reporting Head
Md Adnan**

Senior Business Analyst,
Quince

**Signature of Internship
Coordinator
Prof. Anupama Vijaykumar**

Dept. of AI & ML
DSCE

**Signature of HoD
Dr. Vindhya P Malagi**

Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Nisarga B**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **Quince**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:

USN: 1DS21AI037

Place: Bangalore

NAME: Nisarga B

OFFER LETTER

Quince

Last Brand Services India Private Limited

CIN : U74999TG2019FTCL34961

Mytri Square, 2-41/11, 6/2, Gachibowli Miyapur Road, Prashanth Nagar Colony, Hyderabad, Telangana, 500086

February 4th, 2025

Dear Nisarga B,

We are happy to extend you a consulting offer to join Quince. As you know, Quince's mission is to democratize access to the world's luxury goods and improve the quality of life of millions. With a relentless focus on the customer, you will be playing a significant role in bringing this mission to life. Additionally, it's important to note that Quince is a Values First organization (see Appendix A). We expect and welcome you to contribute to building a Company that supports the mission and values of the business.

The key elements of your compensation package and the other conditions of your employment are as follows:

Title: Data Intern (Consultant)

Start Date: February 6th, 2025

Location: Bangalore, India

Consultant fee: 27500/- per month

Note : The duration of this contract is for Three-months.

Position and Duties: You shall serve in the position of Data Intern shall perform all the duties of that position as described in **Appendix B**. Your position, job description, salary, duties and responsibilities may be modified from time to time in the sole discretion of the Company. You agree to devote your full business time, attention and best efforts to the business of the Company during the employment relationship. Employment with the Company may be terminated with a notice period of 30 days at the will of the Company. Terms and conditions of employment with the Company may be modified at the sole discretion of the Company with or without cause and with or without notice.

Nisarga, we are really looking forward to making you a part of Quince and taking it into its full potential together.

Accepted by:

Yogesh Sharma

Yogesh Sharma

Nisarga B
Nisarga B

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr Md Adnan , Senior Business Analyst, Quince**, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

NISARGA B [1DS21AI037]

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CHAPTER 1

INTRODUCTION

Data automation and workflow management play a crucial role in improving operational efficiency within fast-paced business environments. During my internship at **Quince**, I had the opportunity to contribute to key automation initiatives within the **Product and Sourcing team**, focusing on streamlining communication, optimizing data flow, and enhancing visibility across teams. The goal was to reduce manual efforts, improve response times to delays, and support better decision-making through real-time updates and accurate data handling.

By leveraging **Google Apps Script**, I developed tools that connected internal tracking systems with **Slack**, enabling instant notifications for delayed workflows. This integration allowed team members to take immediate action, reducing potential bottlenecks and improving overall workflow efficiency. I also supported product catalog updates and created structured **Google Sheets dashboards** that segmented data based on divisions, helping stakeholders quickly access relevant information.

I utilized **Looker** to write simple queries that pulled pending workflow data, providing the team with actionable insights to manage operations more effectively.

Key benefits of the automation work included:

- **Real-Time Alerts:** Immediate Slack notifications for delayed tasks ensured faster responses and proactive issue resolution.
- **Reduced Manual Effort:** Automated data handling and dashboard generation minimized repetitive tasks and human error.
- **Improved Organization:** Creating separate sheets for different divisions brought clarity to complex datasets and helped teams focus on their specific areas.
- **Enhanced Decision-Making:** Access to updated and structured data supported timely and informed operational decisions.

In conclusion, my internship at Quince allowed me to apply practical data skills in a real-world setting, where automation and efficient workflow design significantly impacted team productivity. Through this experience, I gained valuable insight into the importance of data-driven processes in modern business operations and developed a strong foundation for future roles in data and process optimization.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

QUINCE



Quince is a rapidly growing, direct-to-consumer lifestyle brand that focuses on delivering high-quality essentials across apparel, home, travel, and accessories at radically lower prices. Founded with a mission to democratize luxury, Quince aims to bridge the gap between quality and affordability by removing traditional retail markups and streamlining supply chains.

Quince operates on a manufacturer-to-consumer model, which cuts out middlemen, allowing them to offer premium products—such as cashmere sweaters, silk shirts, organic bedding, and leather goods—at a fraction of the price typically seen in the retail market. The company emphasizes transparency, sustainability, and ethical sourcing, working closely with trusted manufacturers across the globe to ensure responsible production practices.

With a strong data-driven approach, Quince heavily relies on automation, analytics, and process optimization to support its operations across sourcing, product development, logistics, and customer experience. Teams are empowered to leverage technology to improve efficiency, scale operations, and create a seamless experience for both vendors and customers.

Headquartered in San Francisco, California, Quince has embraced a hybrid and globally distributed team culture, fostering collaboration between cross-functional departments including Product, Sourcing, Engineering, Marketing, and Operations.

Quince

VISION STATEMENT

To create a future where high-quality products are accessible to all by reimagining the way luxury is made, sold, and experienced.

MISSION STATEMENT

To democratize luxury by offering sustainably made, premium products at affordable prices through a tech-enabled, manufacturer-to-consumer model.

CORE VALUES

- **Quality First:** Focus on craftsmanship, durability, and timeless design.
- **Radical Transparency:** Openly share pricing, sourcing, and production details.
- **Customer-Centric:** Build with the customer in mind at every touchpoint.
- **Sustainability:** Prioritize eco-friendly materials and responsible production.
- **Innovation:** Embrace technology and automation to improve efficiency and scale responsibly.

RECENT ACHIEVEMENTS

- **Rapid Growth:** Quince experienced exponential growth in online sales and customer base within just a few years of launch, emerging as a disruptor in the direct-to-consumer retail space.
- **Sustainable Expansion:** Increased its catalog of sustainable products, including organic cotton apparel, recycled polyester outerwear, and ethically sourced leather goods.
- **Technology Integration:** Enhanced internal automation and supply chain systems, enabling faster product updates, smarter inventory management, and real-time workflow tracking.
- **Positive Press & Recognition:** Featured in leading publications such as *Forbes*, *Vogue*, and *Fast Company* for its transparent pricing, minimalist branding, and commitment to sustainability.

FOUNDER PROFILE

Sid Gupta, the founder and CEO of Quince, is a visionary entrepreneur with a strong background in e-commerce and operations. Prior to founding Quince, Sid held leadership positions at companies like McKinsey & Company and Dropcam, and he brings a unique mix of operational discipline and customer empathy to the brand.

Driven by the belief that luxury shouldn't be a privilege, Sid built Quince to challenge traditional retail models and create a more inclusive, tech-enabled shopping experience. Under his leadership, Quince has scaled rapidly while staying true to its core values of quality, transparency, and sustainability. His commitment to innovation and ethical business practices continues to steer Quince toward becoming a global leader in affordable luxury.



CHAPTER 3

SCOPE OF WORK

During my internship at Quince, I worked on a range of data-focused and automation-driven projects within the Product and Sourcing team, aimed at improving operational efficiency, communication, and workflow visibility. My contributions spanned across data collection, product catalog updates, and the automation of real-time notifications for delayed workflows. These tasks were central to streamlining the end-to-end product lifecycle, from development and sourcing to cross-functional tracking and team coordination.

Project 1: Workflow Data Collection and Product Catalog Management

Objective:

To support accurate product lifecycle tracking by collecting structured data from multiple sources and updating the centralized product catalog used for monitoring workflow stages.

Scope:

- Collected product-related data from various internal sources, ensuring consistency and completeness across multiple product attributes.
- Updated the master product catalog in Google Sheets, adding key information such as product names, styles, owners, launch dates, and current workflow statuses.
- Identified and corrected data inconsistencies, missing entries, and mislabeling to maintain high-quality datasets for automation and reporting.
- Ensured that the catalog remained synchronized with live updates from sourcing and production teams.

Impact:

By improving the accuracy and completeness of the product catalog, this work laid the foundation for downstream automation, Slack integration, and reliable workflow tracking. It ensured that automated systems had dependable input data and that stakeholders could make informed decisions based on real-time catalog insights.

Project 2: Slack Workflow Automation System

Objective:

To build an automated system that reads data from the product tracking sheet and sends formatted Slack notifications for workflows that are delayed, with direct action capabilities built into Slack.

Scope:

- Developed a Google Apps Script (testrun_1) that scans the master sheet, identifies late workflows, and sends structured Slack messages using the Slack API.
- Implemented interactive buttons within Slack that allow product owners to update workflow status directly without returning to the spreadsheet.
- Built logic for batching messages to comply with Slack's block limitations and structured dynamic formatting for each notification.
- Included timestamps and action logs to track user engagement and improve accountability.

Impact:

This system significantly reduced manual follow-ups and allowed for real-time visibility of delays, improving cross-functional collaboration and enabling faster resolution of bottlenecks.

Project 3: Division-Based Workflow Sheet Generator

Objective:

To simplify data visibility for individual teams by automatically segmenting the master product workflow sheet into separate sheets based on product divisions.

Scope:

- Built a script to generate division-specific sheets (e.g., Men's, Women's, Home, Accessories), filtering and formatting data based on pre-set criteria.
- Ensured that the sheets were auto-updated as new data was added to the master tracker.

- Included data validation and exception handling to deal with incomplete or unexpected entries.

Impact:

This feature allowed teams to focus only on their relevant data, improving efficiency and reducing the time spent navigating a shared sheet.

This internship allowed me to work across the full spectrum of data handling, automation, and communication tools, enhancing my technical skills in scripting, workflow design, and live system integration. I gained valuable exposure to working in a high-growth, cross-functional environment where data accuracy, speed, and collaboration are key drivers of success.

CHAPTER 4

LEARNING OBJECTIVES

Mastery of Data Automation and Workflow Integration

Develop strong proficiency in data automation techniques and real-time workflow integration by leveraging tools such as Google Apps Script, Google Sheets, and Slack API. Interns are expected to independently acquire skills in scripting custom automation pipelines that reduce manual workload and ensure timely communication of critical workflow updates. Through hands-on experience, interns will demonstrate the ability to automate repetitive processes, integrate multi-platform systems, and manage asynchronous communication triggers, enabling them to deliver robust and scalable workflow automation solutions.

Proficiency in Collaborative Tools and Communication APIs

Gain advanced knowledge and practical skills in working with collaborative tools and APIs, with a focus on Slack interactivity, webhooks, and structured JSON-based messaging. Interns are expected to independently explore and implement Slack's interactive components (such as buttons and block kits) and build full-stack integrations that allow two-way communication between Slack and Google Sheets. This includes handling HTTP requests, writing payload handlers, and implementing real-time response systems. Mastery in this area empowers interns to streamline team collaboration and enhance responsiveness in dynamic operational environments.

Advanced Spreadsheet Programming and Data Management

Develop expertise in advanced spreadsheet programming using Google Apps Script, including writing efficient scripts for data filtering, dynamic sheet generation, and conditional formatting. Interns are expected to automate the creation of segmented datasets (e.g., by division or owner) and ensure real-time syncing between different data views. Through self-directed scripting and validation logic, interns will build systems that enhance data accessibility, improve structure, and ensure high-quality information management across cross-functional teams.

Practical Understanding of Product Lifecycle and Workflow Stages

Acquire a strong operational understanding of the product lifecycle, including key workflow stages, status tracking, and ownership handoffs in a retail/e-commerce setting. Interns are expected to learn how sourcing, production, and product teams collaborate to move items through defined checkpoints. By participating in product catalog updates and workflow tracking, interns will develop insights into the

complexity of cross-functional operations and how data can be leveraged to identify delays, bottlenecks, and improvement opportunities.

Competency in Real-Time Systems Monitoring and Feedback Loops

Gain hands-on experience in building and maintaining real-time feedback systems that support accountability and visibility into ongoing workflows. Interns will learn to develop monitoring tools that collect, update, and track user actions—such as Slack button interactions—and reflect these in live dashboards. This objective equips interns with the knowledge to build closed-loop systems for task tracking, progress monitoring, and automated reporting.

Application of Technical Debugging and Error-Handling Strategies

Build skills in identifying, debugging, and resolving technical issues in live systems. Interns will gain exposure to debugging Apps Script errors, parsing webhook payloads, and troubleshooting API limitations such as Slack's message block constraints. Through trial and error, test cases, and system logs, interns will cultivate a methodical approach to troubleshooting, enhancing the reliability and maintainability of deployed solutions.

Exposure to Cross-Functional Collaboration and Communication

Understand and participate in collaborative workflows with cross-functional teams, including sourcing, operations, and product development. Interns are expected to attend regular team syncs, provide updates on automation tools, and receive live feedback for iterative improvements. This objective emphasizes communication, adaptability, and the importance of aligning technical tools with business needs.

CHAPTER 5

CHALLENGES AND SOLUTION

1: Price Collection from Multiple Competitor Websites and Price Matching

Problem:

Collecting accurate and up-to-date pricing data from multiple competitor websites posed several challenges:

- **Different Pricing Formats:** Competitors displayed prices in various formats, including different currencies, symbols, and even dynamic pricing models (e.g., discounts, regional pricing).
- **Inconsistent Product Listings:** The same product may appear under slightly different names or product descriptions across different websites, making it challenging to map the prices correctly.
- **Frequent Price Fluctuations:** Prices on competitor websites can change frequently, sometimes even multiple times per day, which required real-time tracking for accurate comparison and matching.
- **Hidden Prices:** Some competitors might hide the actual price until a user is logged in, or it may require additional steps (e.g., selecting shipping options or entering location) to see the final price.

Competitor websites may block or restrict access based on geographic IP addresses, making it difficult to scrape data consistently from global sources.

Solution with ExpressVPN:

To mitigate this issue, I used ExpressVPN to access competitor websites from different regions and countries. This allowed me to:

- **Bypass Geo-Restrictions:** Many websites provide different prices or content based on the user's location (e.g., price differences based on the country). By using ExpressVPN, I was able to change the IP address and virtually “relocate” to different countries, thereby simulating visits from those regions. This enabled access to region-specific prices and content.
- **Prevent IP Blocking:** When scraping a competitor website repeatedly from the same IP address, there's a risk of being blocked or throttled due to high request rates. Using ExpressVPN, I could rotate IP addresses, reducing the risk of triggering anti-bot measures and ensuring uninterrupted data collection.

- **Access Regional Product Listings:** Some competitor websites show region-specific offers or products based on the country from which the request originates. ExpressVPN enabled me to choose specific server locations that corresponded to the target regions, ensuring that I gathered relevant, location-specific price data.

2. Product Matching and Price Mapping.

"Men's Running Shoes" on one site might be listed as "Running Sneakers for Men" on another. The fuzzy matching system would link these two as the same product based on similar characteristics, ensuring accurate price comparison.

3: Managing Large-Scale Data Across Multiple Teams

Problem:

One of the major challenges was dealing with large datasets across multiple divisions (product, sourcing, etc.) and ensuring that the data was updated consistently across all systems. Manually tracking and updating product information in Google Sheets created significant inefficiencies, and it was prone to human error.

Solution:

To solve this, we built an automated system using Google Apps Script that allowed real-time updates to product information in Google Sheets based on Slack feedback and workflow progress. We implemented conditional formatting and data validation rules to ensure that data entered into sheets was accurate. We created different sheet views automatically segmented by divisions or workflow stages, allowing different teams to work in parallel without risking data overlap.

4: Ensuring Real-Time Communication Across Platforms

Problem:

The communication of delayed workflows and updates between product tracking systems and Slack was not real-time, leading to miscommunication and inefficient follow-ups. There was a lack of immediate notifications about product delays, which resulted in slower decision-making processes.

Solution:

To resolve this, we developed a Slack notification system using Google Apps Script that provided real-time updates to Slack whenever there were delayed workflows or required actions. This system automatically retrieved the product data from Google Sheets and sent notifications directly to specific Slack channels. The notifications were designed with interactive buttons to enable teams to take immediate action within Slack, further improving communication efficiency.

5: Handling API Limitations and Constraints

Problem:

The Slack API has strict limits on the size of messages and blocks that can be sent. This created challenges when trying to send large amounts of product data or workflow statuses to Slack, as these exceeded the message size limits and caused failures in communication.

Solution:

To overcome this, we implemented a batch processing system that broke down large messages into smaller chunks that adhered to Slack's message size limits. We also used Slack's Block Kit to format the messages, ensuring that they were visually appealing and informative without overwhelming users. This made the communication process more efficient and streamlined, while also ensuring that the data was appropriately presented and actionable.

6: Handling Real-Time Interactions and Updating Data

Problem:

One of the most critical tasks was updating Google Sheets in real-time based on the interaction with the Slack buttons. This required handling multiple interactions simultaneously, ensuring that data was updated correctly and that the system was responsive.

Solution:

We tackled this by creating a custom webhook handler using the `doPost()` function in Google Apps Script, which captured real-time Slack button presses and sent updates back to the corresponding rows in Google Sheets. I used timestamping to log changes and ensure that each update was tracked accurately. We built an efficient error-handling system that caught any inconsistencies during button clicks, making the system more resilient.

HARDWARE AND SOFTWARE REQUIREMENTS

Software Requirements:

- **Operating System:** Windows 10 64-bit
- **Google Apps Script:** (Web-based, no installation required)
- **Microsoft Excel:** Version 2019 or later (or Microsoft 365)
- **Looker:** (Web-based, no installation required)

Hardware Requirements:

- **Processor:** Intel i7 or equivalent
- **RAM:** 8 GB minimum
- **Display:** 23-inch LED Display (Full HD)
- **Input Devices:** Mouse, Keyboard
- **LAN Connection Port:** Required for network connectivity
- **Internet Connection:** Fiber Optic Cable for high-speed internet access

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

1. Competitor Price Data Collection and Real-time Updates:

- Objective: Collect competitor pricing data and ensure the product catalog is continuously updated for competitive pricing analysis.
- Contribution: Implemented a custom price scraping system to gather competitor pricing data from multiple sources, enabling real-time price comparison and updates to internal product catalogs.
- Achievement: Automated the data collection and catalog updating process, ensuring the company always had the most up-to-date competitor pricing information available for analysis.

2. Creation of Dynamic Sheets Based on Divisions:

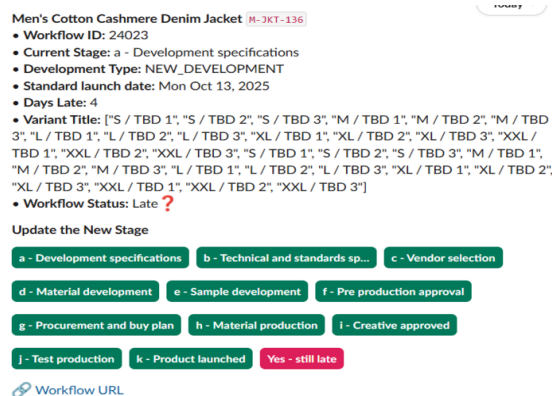
- Objective: Streamline the management of product data by segmenting it into different divisions, improving the organization and accessibility of information.
- Contribution: Designed an automated system using Google Apps Script that dynamically created separate sheets for different divisions from a single master sheet, making it easier to manage and navigate large amounts of data.
- Achievement: Enhanced data organization by creating division-specific sheets from a central master sheet, which simplified reporting and made it easier for teams to access relevant data specific to their needs.

3. Real-time Notifications via Slack for Workflow Delays:

- Objective: Ensure teams are promptly informed about any delays in product catalog updates to avoid bottlenecks.
- Contribution: Integrated Slack notifications into the workflow, allowing team members to receive real-time alerts for any workflows that were not updated on time.
- Achievement: Enabled a more responsive workflow process by using real-time notifications, improving the team's ability to quickly address any issues related to delayed product catalog updates, ensuring smoother operations and minimizing any potential delays in catalog updates.

4. Improvement in Team Collaboration and Productivity:

- Objective: Enhance the productivity and collaboration of teams handling product catalog updates and competitor price monitoring.
- Contribution: By automating manual tasks (e.g., updating workflows and sending notifications) and organizing the data into clear, manageable divisions, streamlined the workflow process and reduced redundant work.
- Achievement: Boosted team collaboration and productivity by eliminating time-consuming manual tasks and ensuring each division could work with the most relevant and up-to-date data, leading to quicker decision-making and more accurate competitor analysis.



5. Simplified Reporting and Data Management:

- Objective: Improve the efficiency of reporting and data tracking for different product divisions.
- Contribution: Created automated reports based on specific product divisions, which provided real-time insights into the product catalog and competitor price data.
- Achievement: Simplified reporting across the organization by categorizing data into individual division-based sheets, which allowed for easier and faster access to relevant data, leading to more informed decisions across different business units.

6. Optimization of Catalog Management Workflows:

- Objective: Optimize product catalog management to ensure it reflects accurate, up-to-date competitor pricing and internal catalog updates.

- Contribution: Automated the product catalog update workflows and incorporated a real-time tracking system that alerted team members if any catalog updates were missing or delayed.
- Achievement: Optimized catalog management by automating tasks, improving efficiency, and ensuring that the catalog remained current and competitive in the market.

CHAPTER 7

REFLECTION AND ANALYSIS

Throughout this project, I gained valuable insights into data management, automation, and cross-functional workflows. By working on competitor price data collection and developing solutions to optimize workflows in our product catalog, I was exposed to several technical challenges that helped me refine my skills in data handling, automation, and collaboration.

One of the most critical aspects of the project was the collection and aggregation of competitor price data. To automate this process, I employed custom web scraping techniques that allowed us to pull real-time price data from various competitor websites. This exercise sharpened my skills in data extraction, where I learned how to handle dynamic content, deal with inconsistent data structures, and ensure accuracy and consistency across different sources. The experience deepened my understanding of how real-time data collection can be integrated into business operations, providing actionable insights into the competitive landscape and helping inform our pricing strategies.

Another key area of growth came from automating workflows for the product catalog using Google Apps Script. By leveraging this tool, I was able to streamline repetitive tasks, such as sending late workflow notifications through Slack and managing product data updates efficiently. Through this experience, I not only enhanced my technical proficiency in script automation but also gained a deeper understanding of how automation can improve operational efficiency by reducing human intervention and the potential for errors. Implementing automated workflows ensured that the latest product data was always up to date, which ultimately contributed to better decision-making and optimized performance in product catalog management.

I worked on data organization, specifically by creating separate sheets based on different divisions from a single sheet. This was a crucial step in improving the accessibility and usability of our data. By organizing and segmenting the data into logical groupings, I made it easier for teams to focus on their specific areas of interest without sifting through unnecessary information. This also highlighted the importance of data organization and structure when managing large datasets, as it directly impacts the ability to extract meaningful insights and make data-driven decisions.

Throughout this project, I learned the significance of collaboration between technical and non-technical teams. The successful implementation of the solution required a thorough understanding of the operational requirements and workflows, and frequent communication with stakeholders was key to ensuring alignment between our objectives and their needs.

Cross-functional collaboration enhanced my ability to explain technical processes to non-technical team members, making it easier to adopt and scale automation processes across departments.

This project not only sharpened my technical skills in automation, data collection but also emphasized the broader impact of these solutions on business operations. It reinforced the importance of efficiency and accuracy in data management while highlighting how even small optimizations can significantly enhance team productivity and decision-making.

In conclusion, this project provided me with an excellent opportunity to apply my technical expertise in real-world scenarios while also enhancing my skills in automation and collaboration. The experience has broadened my understanding of how technology can be leveraged to improve business processes and workflows, and I look forward to applying these insights to future projects.

CHAPTER 8

RECOMMENDATIONS

1. **Implement Structured Programs:** To ensure a comprehensive and impactful internship experience, it is recommended to develop structured programs with clearly defined learning objectives, project assignments, and mentorship opportunities. These programs should include a balance of classroom training, hands-on projects, and regular mentorship sessions. Establish clear milestones and deliverables to guide interns' progress, providing a well-rounded roadmap for their learning journey. Regular assessments can help track their development and ensure the program meets both the interns' and the organization's goals.
2. **Foster Collaboration with Educational Institutions:** Building partnerships with educational institutions is essential for strengthening the program and tapping into fresh talent. Establish joint internship programs, research collaborations, and academic exchanges to bridge the gap between academia and industry. Collaborate with faculty members and research centers to offer internships tied to ongoing research projects, providing students with real-world applications of their academic work. This not only enriches the interns' experience but also contributes valuable insights to the organization from cutting-edge academic research.
3. **Provide Mentorship and Support:** Mentorship plays a crucial role in the development of interns. Pair each intern with an experienced mentor who can provide guidance, feedback, and support. Mentors can help interns navigate challenges, develop key skills, and achieve personal and professional goals. In addition to one-on-one mentorship, regular check-in meetings should be scheduled to discuss progress, address concerns, and provide feedback. Offering mentorship workshops can further enhance the quality of guidance provided, ensuring that mentors are equipped to maximize their impact.
4. **Facilitate Networking Opportunities:** To enhance the interns' professional development, networking opportunities should be a priority. Organize meet-and-greet sessions, panel discussions, and industry tours where interns can connect with professionals, employees, and industry experts. Establishing online networking platforms will also ensure that interns have a continued space to connect even after the internship ends. This will help build lasting relationships and expose them to a broader range of career paths, fostering collaboration and future professional opportunities.

CHAPTER 9

CONCLUSION

In conclusion, the work completed throughout this project has significantly contributed to improving data handling, workflow automation, and overall efficiency in monitoring and updating the product catalog. The collection and processing of competitor price data through web scraping enabled a more competitive and informed pricing strategy, providing valuable insights into market trends and aiding in price adjustments across various product categories. This data-driven approach has enhanced the organization's ability to remain competitive in a dynamic market.

The automation of workflows within the product catalog using AppScript has not only streamlined the process but also ensured that updates and changes are made in real-time. This automation, combined with the use of Slack for real-time alerts, ensured that no delayed workflows went unnoticed, minimizing operational inefficiencies and providing quick resolutions. Moreover, the ability to segment and organize the product catalog data into different sheets based on various divisions allowed for better management and a more organized workflow.

Through these improvements, I have gained valuable experience in applying scripting and automation to real-world problems, while also learning how to handle large datasets and create efficient processes that save time and improve accuracy. The integration of these technologies into the organization's workflow has had a direct impact on productivity, making tasks more manageable and ensuring seamless collaboration across teams.

Ultimately, this project has provided me with a solid foundation in automating business processes, analyzing market data, and developing efficient data management strategies. The skills gained here will undoubtedly be applicable in future roles, as the ability to work with complex data and improve operational workflows is critical in today's fast-paced business environment. The work completed has not only improved the efficiency of the product catalog management but also created a more responsive and agile team that can adapt to new challenges and market changes.

CHAPTER 10

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7. <https://www.youtube.com/playlist?list=PL2rFVcDw2yVoXII3onYS7Ecufjg2ODI9u>